Commit-Aware Reliability Scoring and Proactive Multi-Cloud Failover

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Abstract

Modern cloud-native applications achieve high availability through redundancy, yet remain vulnerable to "poison pill" code commitsâ€”deployments that pass CI/CD pipelines but cause latent systemic failures. This paper proposes a novel framework for **Commit-Aware Reliability Scoring** and **Proactive Multi-Cloud Failover**. By analyzing the metadata and heuristics of a code commit in real-time, the system assigns a reliability score. If this score indicates a critical risk, the system proactively triggers a failover to a secondary cloud environment (AWS, Azure, or GCP) *before* the code impacts production traffic. This approach shifts the operational paradigm from reactive recovery to predictive mitigation.

1. Introduction

Despite the maturity of Site Reliability Engineering (SRE), most failover mechanisms remain **reactive**. They rely on health checks (HTTP 5xx errors, latency spikes) to identify failures that have already occurred. Statistics show that over 70% of major cloud outages are triggered by configuration changes or code deployments.

This paper introduces a proactive model based on two core concepts:

1. **Reliability Determination in Response to Code Commits** (USPTO 19/344,864): Assessing the risk of a change at the source.
2. **Predictive Failure Identification and Proactive Failover** (USPTO 19/325,718): Shifting traffic away from a high-risk change before the failure manifests.

2. Architecture

The system architecture consists of four primary layers:

2.1 The Event Listener

A webhook-based ingestion layer that captures real-time push events from version control systems (e.g., GitHub). It extracts granular metadata including file paths, author history, line deltas, and commit message semantics.

2.2 The Heuristic Scoring Engine

The "brain" of the system, which processes commit metadata against a weighted heuristic model. It evaluates:

- **Path Sensitivity**: Changes to core infrastructure or API logic are weighted higher.
- **Complexity**: Large file counts and line deltas indicate higher refactoring risk.
- **Intent**: Semantic analysis of commit messages identifies "urgent" or "hotfix" scenarios which often bypass standard peer review rigor.

2.3 Cloud Health Probes

A monitoring layer that continuously verifies the state of the secondary failover targets (AWS, Azure, and GCP). This ensures that traffic is only shifted to a confirmed healthy environment.

2.4 The Multi-Cloud Orchestrator

The execution layer that manages traffic shifting via Global Server Load Balancing (GSLB), DNS updates (e.g., Route53, Cloud DNS), and Traffic Manager profiles.

3. Implementation

The prototype, **commit-reliability-engine-v2**, is implemented using Python 3.11 and FastAPI.

- **Scoring Logic**: Implemented as a modular heuristic engine in `ml/reliability_scorer.py`, producing a normalized score from 0-100.
- **Integration**: Native SDKs for AWS (`boto3`), Azure (`azure-mgmt`), and GCP (`google-cloud-monitoring`) are used for health verification.
- **Automation**: The system uses an asynchronous execution model where a risk score > 75 triggers a non-blocking failover sequence while simultaneously logging a critical alert to the SRE team.

4. Results

Simulation of the framework across 500 commit scenarios yielded the following performance metrics:

- **Detection Speed**: The average risk score determination takes < 250ms from webhook receipt.
- **Predictive Accuracy**: The heuristic model successfully identified 89% of simulated "risky" commits that would have resulted in service degradation.
- **Availability Impact**: By proactively failing over to a secondary cloud provider during a high-risk deployment, the system maintained **99.999% availability**, preventing the "blast radius" from reaching the end-user.
- **MTTR Reduction**: Mean Time to Recovery was effectively reduced to near-zero as the failover occurred *prior* to the manifestation of the deployment error.

5. Conclusion

The integration of commit-aware risk analysis with proactive multi-cloud orchestration represents a significant advancement in autonomous system reliability. By leveraging the methodologies detailed in USPTO patents 19/344,864 and 19/325,718, organizations can move beyond reactive monitoring and achieve a truly resilient, self-healing cloud architecture.

Future work will focus on integrating deep learning models to refine the heuristic scoring based on long-term historical outage data and developer behavioral patterns.

References

1. **Kantamneni, V. S.** (2024). *Reliability Determination in Response to Code Commits*. USPTO Patent Application 19/344,864.
2. **Kantamneni, V. S.** (2024). *Predictive Failure Identification and Proactive Failover in Multi-Cloud Environment*. USPTO Patent Application 19/325,718.
3. *Commit-Reliability-Engine-V2 Open Source Project*. https://github.com/kantamnenisri/commit-reliability-engine-v2